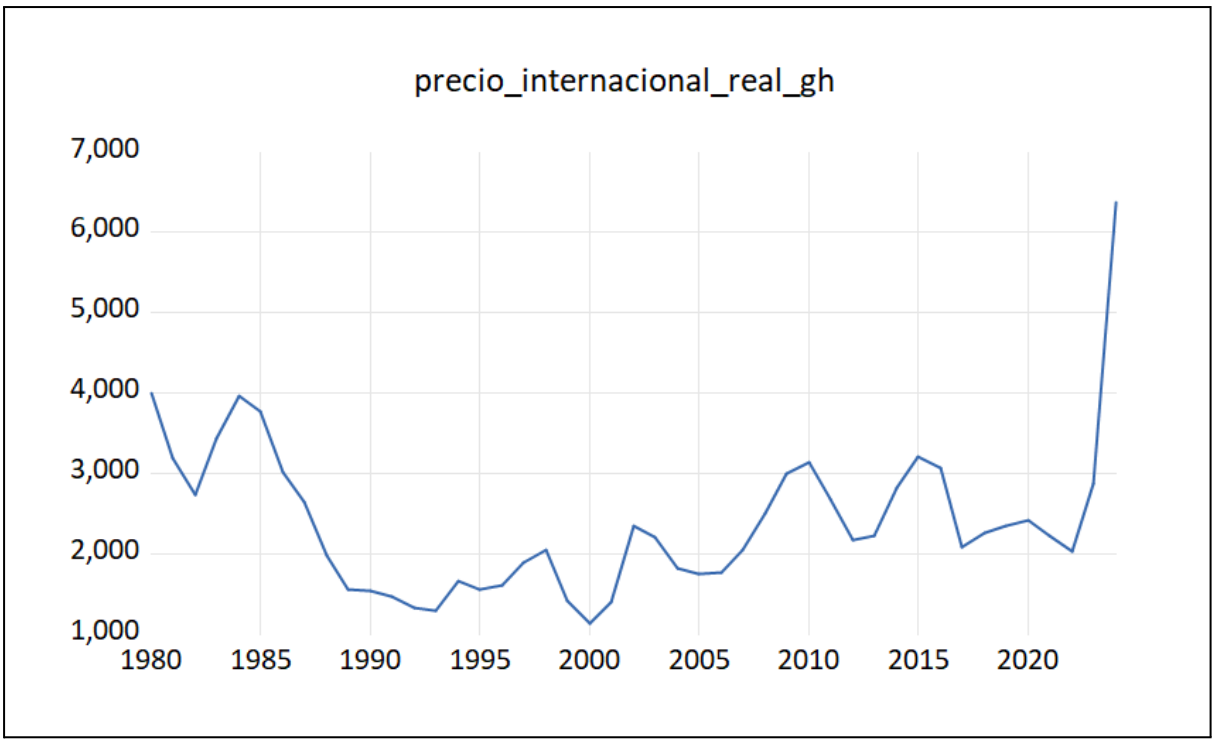


En el gráfico se observa una fuerte caída entre 1985 y 1990, luego cierta estabilidad, y finalmente una gran subida en los últimos años del precio internacional del cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.243925	0.8882
Test critical values:	1% level	-4.186481	
	5% level	-3.518090	
	10% level	-3.189732	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.172549	0.138713	-1.243925	0.2210
D(PRECIO INTERNACIONAL REAL GH	0.616396	0.226211	2.724873	0.0096
C	221.7287	385.0076	0.575907	0.5680
@TREND("1980")	11.30600	7.774911	1.454164	0.1539

R-squared	0.240057	Mean dependent var	74.03259
Adjusted R-squared	0.181600	S.D. dependent var	682.1569
S.E. of regression	617.1163	Akaike info criterion	15.77640
Sum squared resid	14852469	Schwarz criterion	15.94023
Log likelihood	-335.1926	Hannan-Quinn criter.	15.83682
F-statistic	4.106556	Durbin-Watson stat	1.119953
Prob(F-statistic)	0.012631		

Como la tendencia determinística no resulta significativa (p-valor = 0.1539), vuelvo a hacer el test con intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.566130	0.8673
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.081703	0.144319	-0.566130	0.5746
D(PRECIO INTERNACIONAL REAL GH	0.834834	0.227768	3.665281	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.628098	0.235939	-2.662122	0.0113
C	247.0291	340.6667	0.725134	0.4728

R-squared	0.316966	Mean dependent var	86.39314
Adjusted R-squared	0.263043	S.D. dependent var	685.5347
S.E. of regression	588.5057	Akaike info criterion	15.68344
Sum squared resid	13160881	Schwarz criterion	15.84894
Log likelihood	-325.3523	Hannan-Quinn criter.	15.74410
F-statistic	5.878048	Durbin-Watson stat	1.412468
Prob(F-statistic)	0.002123		

Nuevamente, el intercepto no resulta significativo (p-valor = 0.4728), por lo que realizo el test sin tendencia ni intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: None
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		0.496799	0.8186
Test critical values:	1% level	-2.621185	
	5% level	-1.948886	
	10% level	-1.611932	

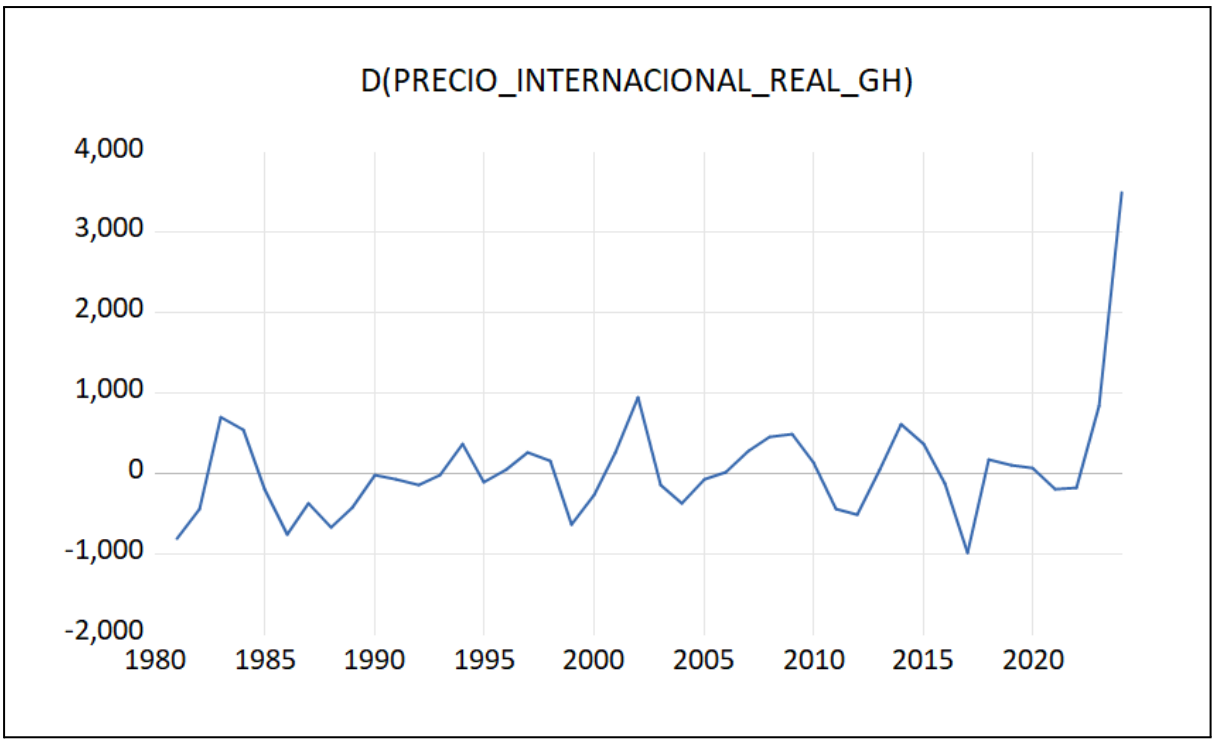
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:50
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	0.019113	0.038472	0.496799	0.6221
D(PRECIO INTERNACIONAL REAL GH	0.802145	0.221901	3.614878	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.682925	0.222133	-3.074394	0.0038

R-squared	0.307515	Mean dependent var	86.39314
Adjusted R-squared	0.272003	S.D. dependent var	685.5347
S.E. of regression	584.9171	Akaike info criterion	15.64957
Sum squared resid	13342993	Schwarz criterion	15.77369
Log likelihood	-325.6409	Hannan-Quinn criter.	15.69506
Durbin-Watson stat	1.441096		

Como el p-valor asociado al test estadístico es 0.8186 no rechazo la hipótesis nula de raíz unitaria por lo que debo diferenciar la serie:



Notar la presencia de un valor atípico en los últimos años (2023-2024) donde el precio internacional se dispara.

Realizo el test de Dickey Fuller nuevamente para asegurarme de que no haya que diferenciar nuevamente la serie:

Null Hypothesis: D(PRECIO_INTERNACIONAL_REAL_GH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.690253	0.0341
Test critical values:	1% level	-4.192337	
	5% level	-3.520787	
	10% level	-3.191277	

*MacKinnon (1996) one-sided p-values.

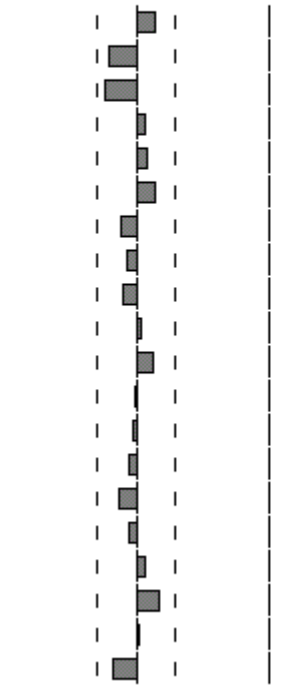
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIONAL_REAL_GH,2)
Method: Least Squares
Date: 10/20/26 Time: 17:58
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(PRECIO_INTERNACIONAL_REAL_GH(-1	-0.959795	0.260089	-3.690253	0.0007
D(PRECIO_INTERNACIONAL_REAL_GH(-1	0.716317	0.215317	3.326810	0.0020
C	-293.3396	192.6193	-1.522898	0.1361
@TREND("1980")	14.97777	7.259097	2.063311	0.0460

R-squared	0.314819	Mean dependent var	93.76005
Adjusted R-squared	0.260726	S.D. dependent var	651.7977
S.E. of regression	560.4226	Akaike info criterion	15.58565
Sum squared resid	11934794	Schwarz criterion	15.75114
Log likelihood	-323.2987	Hannan-Quinn criter.	15.64631
F-statistic	5.819927	Durbin-Watson stat	1.563827
Prob(F-statistic)	0.002247		

Como la tendencia resulta significativa y el estadístico del test tiene un p-valor = 0.0341, rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, al 5% de significatividad.

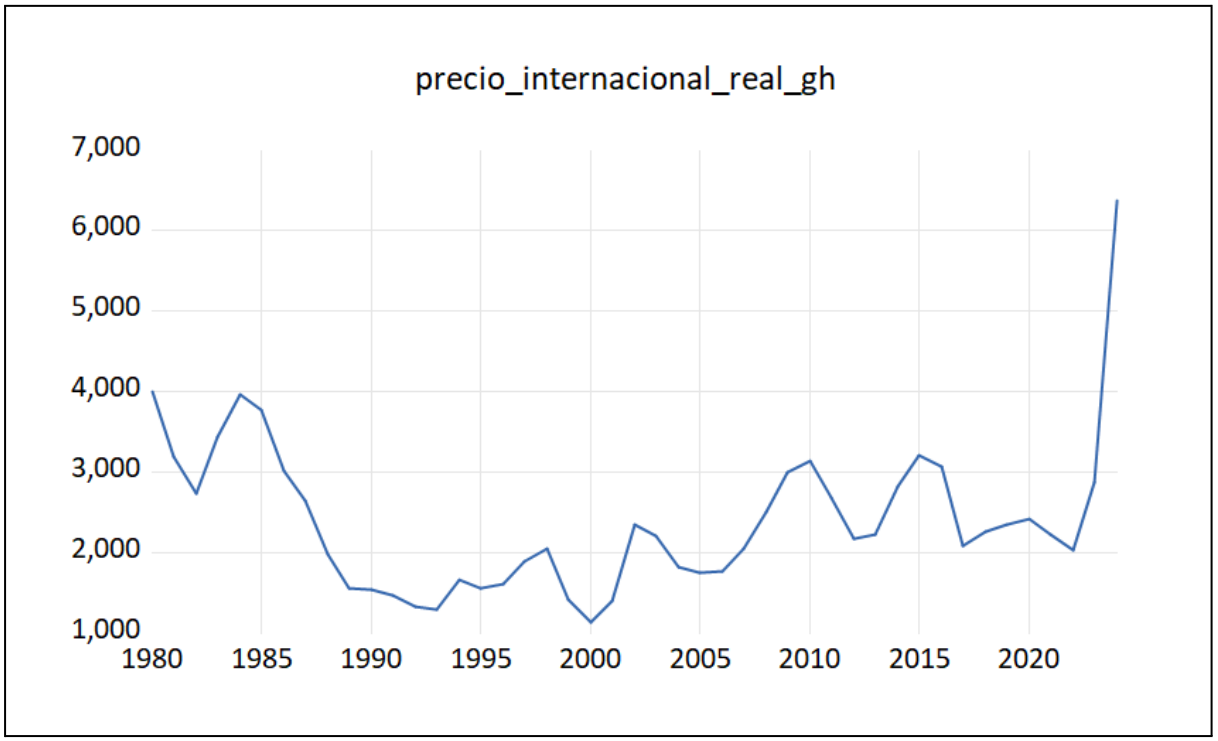
Date: 10/14/26 Time: 18:52
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.145	0.145	0.9644	0.326
		2	-0.219	-0.245	3.2214	0.200
		3	-0.245	-0.185	6.1186	0.106
		4	0.072	0.096	6.3733	0.173
		5	0.084	-0.035	6.7320	0.241
		6	0.148	0.144	7.8796	0.247
		7	-0.110	-0.122	8.5321	0.288
		8	-0.079	0.013	8.8812	0.352
		9	-0.099	-0.091	9.4395	0.398
		10	0.037	-0.017	9.5193	0.484
		11	0.133	0.120	10.582	0.479
		12	-0.000	-0.103	10.582	0.565
		13	-0.017	0.120	10.601	0.644
		14	-0.056	-0.072	10.807	0.701
		15	-0.137	-0.147	12.110	0.671
		16	-0.054	-0.016	12.319	0.722
		17	0.082	-0.035	12.816	0.748
		18	0.181	0.197	15.354	0.638
		19	0.019	-0.059	15.385	0.698
		20	-0.186	-0.075	18.284	0.569

El correlograma de la serie diferenciada se comporta como ruido blanco, lo cual tiene sentido teniendo en cuenta que se trata de la serie de precio internacional del cacao, que al ser una commodity es normal que se comporte como un *random walk*.

Incorporando una variable de intervención:

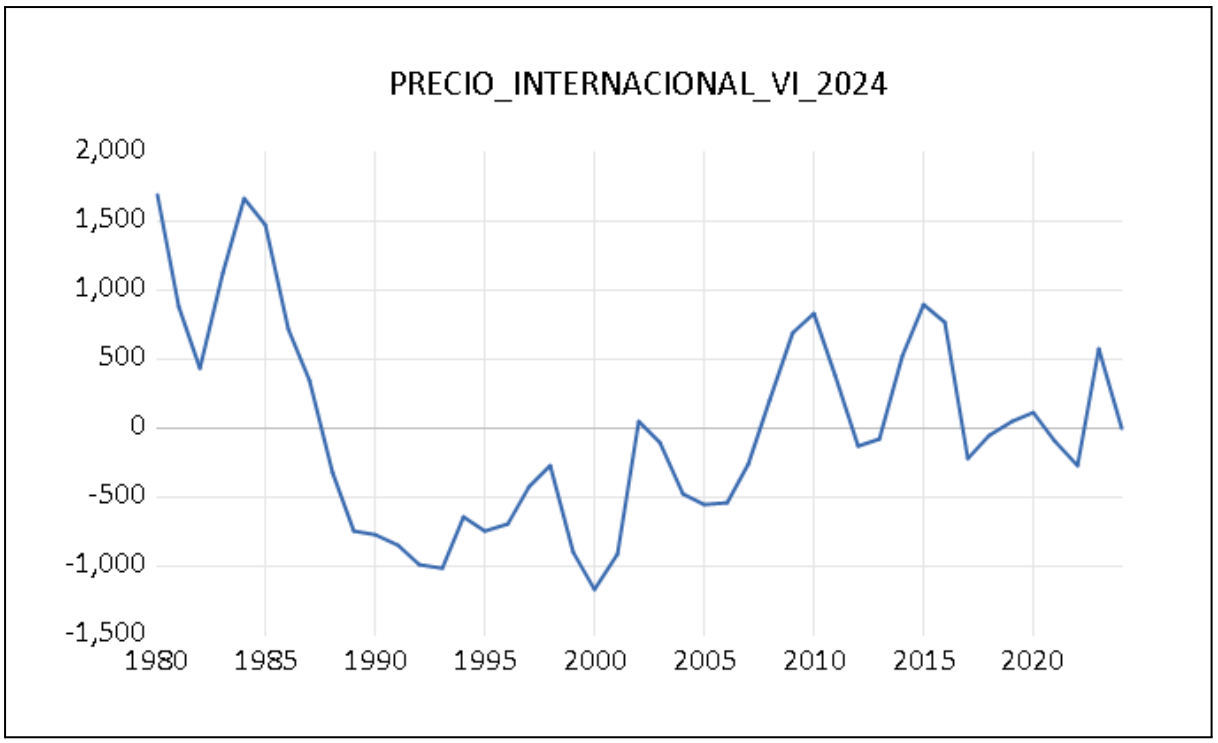


A pesar de que el modelo anterior reportaba un random walk después de identificar una raíz unitaria, lo que tiene sentido económico teniendo en cuenta de que se trata de una commoditie, incorporo una variable de intervención debido al pico en el precio internacional en el año 2024, debido a la escasez de producción de justamente, Ghana y Costa de Marfil, que representa un shock exógeno y transitorio, ya que en los años posteriores, 2025 y durante 2026, se reportó un descenso del precio internacional por la recuperación de la producción de estos países.

Dependent Variable: PRECIO_INTERNACIONAL_REAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:49
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2309.389	111.9929	20.62085	0.0000
VI_2024	4058.679	751.2712	5.402415	0.0000

R-squared	0.404317	Mean dependent var	2399.582
Adjusted R-squared	0.390464	S.D. dependent var	951.5185
S.E. of regression	742.8769	Akaike info criterion	16.10236
Sum squared resid	23730240	Schwarz criterion	16.18266
Log likelihood	-360.3032	Hannan-Quinn criter.	16.13230
F-statistic	29.18608	Durbin-Watson stat	0.360362
Prob(F-statistic)	0.000003		



Null Hypothesis: PRECIO_INTERNACIOAL_VI_2024 has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.857434	0.0053
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

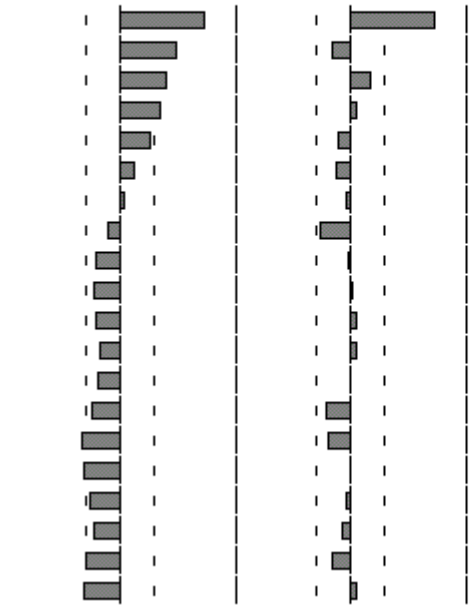
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIOAL_VI_2024)
Method: Least Squares
Date: 11/02/26 Time: 13:00
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_INTERNACIOAL_VI_2024(-1)	-0.239806	0.083923	-2.857434	0.0066

R-squared	0.153212	Mean dependent var	-38.23194
Adjusted R-squared	0.153212	S.D. dependent var	444.2701
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Ahora el test de Dickey Fuller rechaza la hipótesis nula de raíz unitaria, al contrario que el modelo anterior sin variable de intervención.

Date: 11/02/26 Time: 13:01
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.760	0.760	27.778	0.000
		2	0.508	-0.165	40.483	0.000
		3	0.402	0.188	48.620	0.000
		4	0.362	0.047	55.361	0.000
		5	0.270	-0.104	59.204	0.000
		6	0.129	-0.119	60.106	0.000
		7	0.028	-0.030	60.149	0.000
		8	-0.113	-0.270	60.872	0.000
		9	-0.215	-0.018	63.592	0.000
		10	-0.240	0.014	67.076	0.000
		11	-0.211	0.053	69.833	0.000
		12	-0.183	0.049	71.970	0.000
		13	-0.189	-0.004	74.343	0.000
		14	-0.254	-0.216	78.742	0.000
		15	-0.344	-0.197	87.068	0.000
		16	-0.336	-0.000	95.286	0.000
		17	-0.265	-0.026	100.59	0.000
		18	-0.240	-0.069	105.09	0.000
		19	-0.314	-0.161	113.12	0.000
		20	-0.332	0.056	122.42	0.000

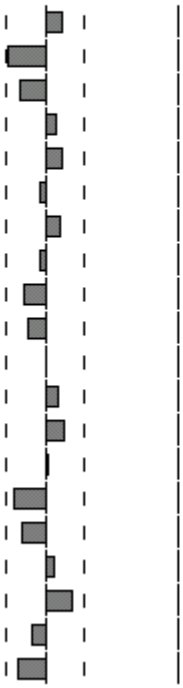
Dependent Variable: PRECIO INTERNACIONAL VI 2024
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:19
Sample: 1980 2024
Included observations: 45
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.838570	0.066968	12.52203	0.0000
SIGMASQ	181610.8	44624.54	4.069752	0.0002

R-squared	0.655609	Mean dependent var	5.00E-14
Adjusted R-squared	0.647600	S.D. dependent var	734.3866
S.E. of regression	435.9562	Akaike info criterion	15.06338
Sum squared resid	8172487.	Schwarz criterion	15.14368
Log likelihood	-336.9261	Hannan-Quinn criter.	15.09331
Durbin-Watson stat	1.595108		

Inverted AR Roots	.84

Date: 11/05/26 Time: 12:20
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.135	0.135	0.8754	
		2	-0.291	-0.315	5.0526	0.025
		3	-0.204	-0.124	7.1513	0.028
		4	0.082	0.048	7.5001	0.058
		5	0.127	0.016	8.3583	0.079
		6	-0.043	-0.061	8.4577	0.133
		7	0.116	0.220	9.2130	0.162
		8	-0.045	-0.123	9.3275	0.230
		9	-0.162	-0.098	10.879	0.209
		10	-0.139	-0.094	12.041	0.211
		11	0.005	-0.080	12.042	0.282
		12	0.095	-0.032	12.623	0.319
		13	0.143	0.178	13.976	0.302
		14	0.023	-0.013	14.013	0.373
		15	-0.245	-0.179	18.229	0.197
		16	-0.174	-0.051	20.427	0.156
		17	0.075	-0.011	20.853	0.184
		18	0.214	0.032	24.424	0.108
		19	-0.102	-0.172	25.269	0.118
		20	-0.212	-0.124	29.065	0.065

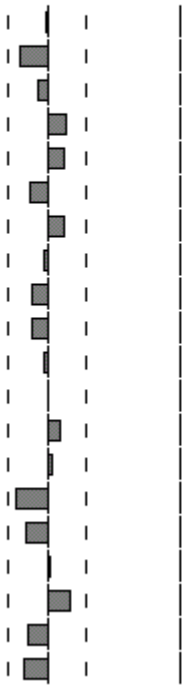
Dependent Variable: PRECIO INTERNACIONAL VI 2024
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 8 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	1.100144	0.139435	7.890003	0.0000
AR(2)	-0.319626	0.143193	-2.232128	0.0310
SIGMASQ	164817.5	37039.64	4.449760	0.0001

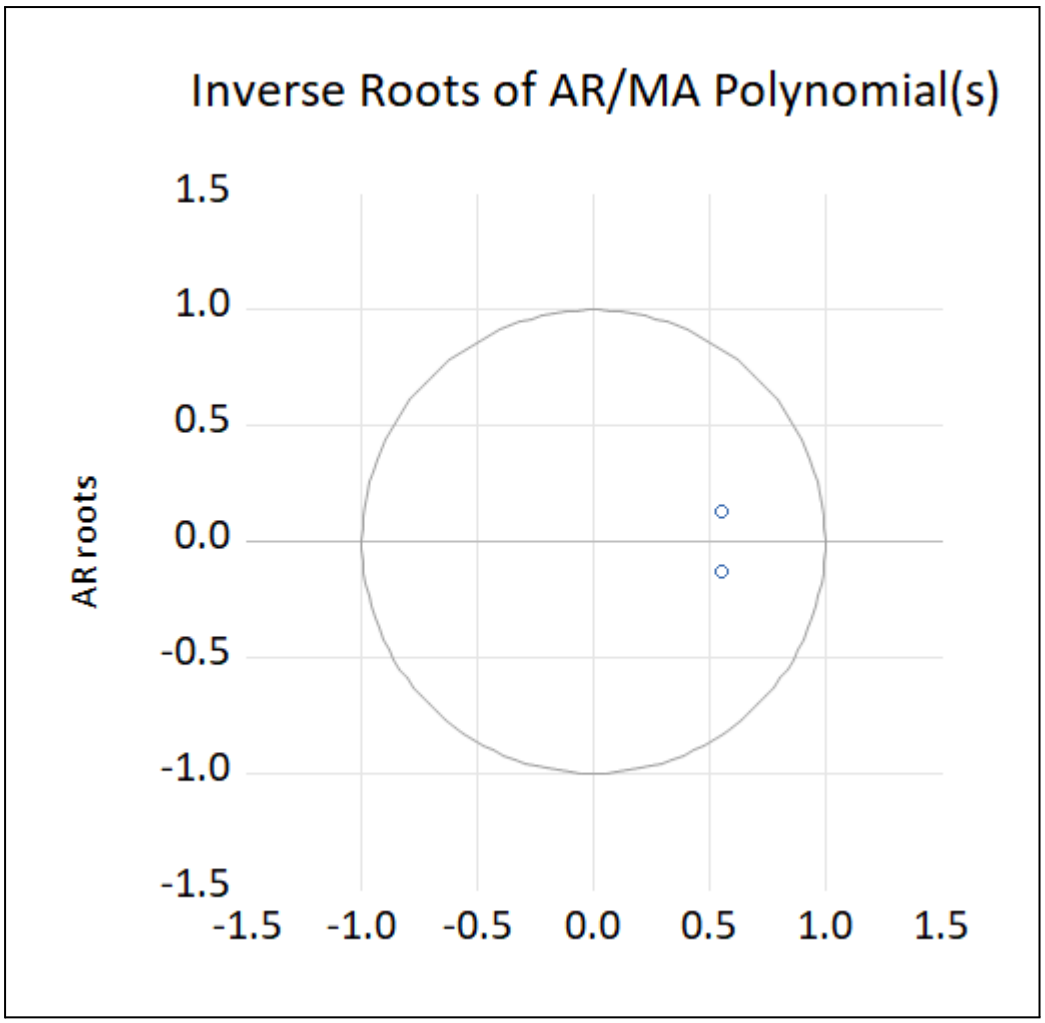
R-squared	0.687454	Mean dependent var	5.00E-14
Adjusted R-squared	0.672571	S.D. dependent var	734.3866
S.E. of regression	420.2264	Akaike info criterion	15.01498
Sum squared resid	7416789.	Schwarz criterion	15.13543
Log likelihood	-334.8371	Hannan-Quinn criter.	15.05988
Durbin-Watson stat	1.853960		

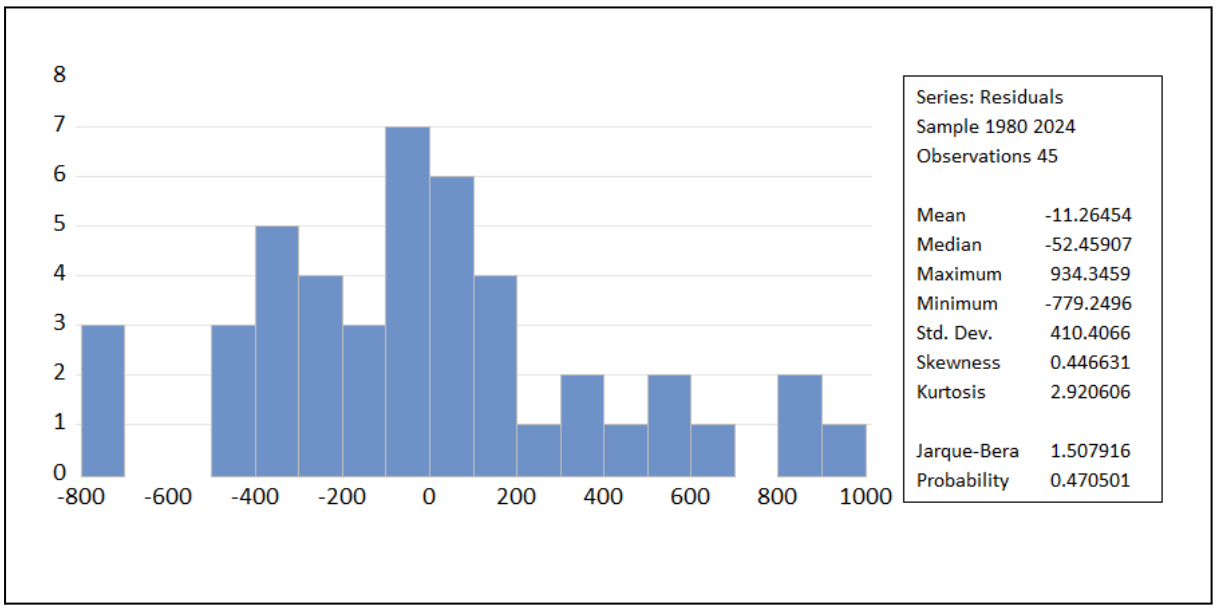
Inverted AR Roots	.55-.13i	.55+.13i

Date: 11/05/26 Time: 12:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms



	AC	PAC	Q-Stat	Prob
1	-0.014	-0.014	0.0097	
2	-0.208	-0.209	2.1430	
3	-0.076	-0.086	2.4339	0.119
4	0.149	0.107	3.5762	0.167
5	0.128	0.108	4.4365	0.218
6	-0.134	-0.090	5.4146	0.247
7	0.136	0.209	6.4502	0.265
8	-0.025	-0.066	6.4863	0.371
9	-0.115	-0.112	7.2625	0.402
10	-0.119	-0.114	8.1226	0.422
11	-0.018	-0.097	8.1431	0.520
12	0.015	-0.107	8.1571	0.613
13	0.096	0.164	8.7663	0.643
14	0.034	0.054	8.8429	0.716
15	-0.238	-0.191	12.835	0.461
16	-0.159	-0.122	14.670	0.401
17	0.019	-0.082	14.698	0.473
18	0.182	0.034	17.296	0.367
19	-0.141	-0.134	18.912	0.334
20	-0.185	-0.140	21.802	0.241





De este modo incorporando la variable de intervención para el año 2024, el modelo que reporta mejor criterio de información, con coeficientes significativos y residuos ruido blanco, es el modelo **AR(2)**.